

new blood vessels that allow tumours to spread. In digestive organs, pollution can affect the gut microbiota and influence the development of cancer.

OTHER AILMENTS

Research on other impacts of air pollution—on the kidney, eyes and diabetes—has just begun. While there is a general understanding of long-term exposure to PM2.5 causing decline in renal health and function, which in turn could trigger age-related macular degeneration (AMD)—a leading cause of blindness—in the eye, the full extent of these effects is just beginning to be acknowledged and understood. "Pollution is not just PM2.5," says Dr Chawla. "Even toxic gases in significant amounts can be carcinogens, inflammatory triggers and attack every organ, including the central nervous system. As air pollution becomes a stark reality for us all, medical research on the subject is growing because we have to understand why, how and how much these pollutants are killing us. There is a series of ongoing research into chronic diseases and their links to air pollution globally as well as in India."

The most recent disorder to be linked with PM2.5, say doctors, is something few expected—diabetes. "This is a surprising new discovery but, yes, air pollution can cause diabetes," says Dr Nangia. A landmark study conducted in India, it was published in the *BMJ Open Diabetes Research and Care* journal. Researchers followed a cohort of 12,000 men and women in Delhi and Chennai from 2010 to 2017 and measured their blood sugar levels periodically. Using satellite data and air pollution exposure models, they determined the air pollution in the locality of each participant in that time-frame. They found that one month of exposure to PM2.5 led to elevated levels of blood sugar, and prolonged exposure of one year or more led to increased risk of diabetes. They found that for every 10 µg/m3 rise in annual average PM2.5 levels in the two cities, the risk for diabetes went up by 22 per cent.

HOW TO STAY SAFE

On days when AQI breaches the severe mark, doctors advise one to stay indoors as much as possible. "Everyone needs to try and maintain indoor air quality by keeping doors and windows shut. Offices should take note and invest in a purification system," says Dr Nangia. Malls, overcrowded spaces and exercising outdoors should also be avoided. Air purifiers may be expensive but can do a world of good to highly vulnerable people (those above 60 or below five). But the room it is being used in must be sealed for it to be properly effective. Dr Chawla, though, is not entirely convinced about the ability of air purifiers to reduce the impact of pollution. "There is little science to show it," he says. "Unless you invest a huge amount and put one in every room." He highly recommends wearing an N95 mask every time you step out instead. This is because staying indoors for months on end is neither healthy nor possible. "I have an elderly parent with respiratory issues and I worry every year when the air quality plummets," he says. "The social isolation of staying locked up is negative for our wellbeing. Masks help one step out."

However, the only real solution is one that cannot happen on an individual level at all. It has to be a public demand for the right to breathe clean air. "European cities would shut down if the AQI was as high as ours. But we still continue to move about. Till we take the threat seriously, it will not go away," says Dr Chawla. Masks and air purifiers help only so much. To be able to live freely and enjoy good long-term health, the quality of air we breathe needs to be significantly better. ■